applications, and is the integration of IoT with the cloud and the World Wide Web for the real-time connectivity of devices.

A number of programming platforms are available for working with IoT, IoE, fog or edge scenarios. Hundreds of toolkits, which are very powerful and easy to use for dynamic research, are available as well. A few of these are listed in Table 1.

Node-RED: A tool for flow based programming of IoT scenarios

Node-RED (*https://nodered.org/*) is a powerful and easy-to-use programming platform for the simulation of IoT scenarios. Fog and edge computing can also be done using flow based programming in Node-RED. Here, high performance structures can be implemented using minimum coding.

Installation and working

Node-RED is a specialised package that is installed onto the Node.js platform. The latter is a lightweight but high performance programming environment based on JavaScript. Many packages are available in Node.js for multiple applications, including Internet of Things (IoT), cloud computing, machine learning, data science, and blockchain.

To work with Node-RED, the Node.js platform should be installed first, which is available at *https://nodejs.org* for multiple operating systems like Windows, Mac and Linux for 32-bit or 64-bit architectures (Figure 1).

After the installation of Node.js, the Node-RED package can be installed from the node package manager (NPM), which is the repository of packages developed and deployed for the Node platform (Figure 2).

In the installation directory of Node.js, the *npm* command is used to install Node-RED, as follows:

```
E:\>cd nodejs
E:\nodejs>npm i node-red
```

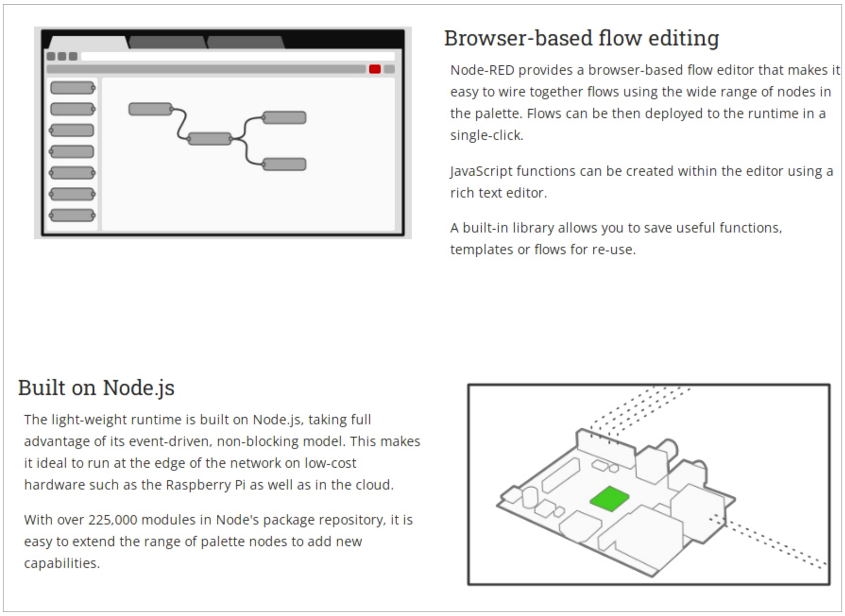


Figure 2: Node-RED platform built on Node.js (Source: *nodered.org*)

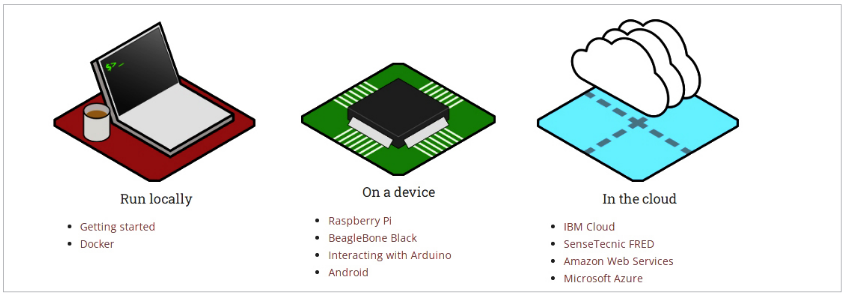
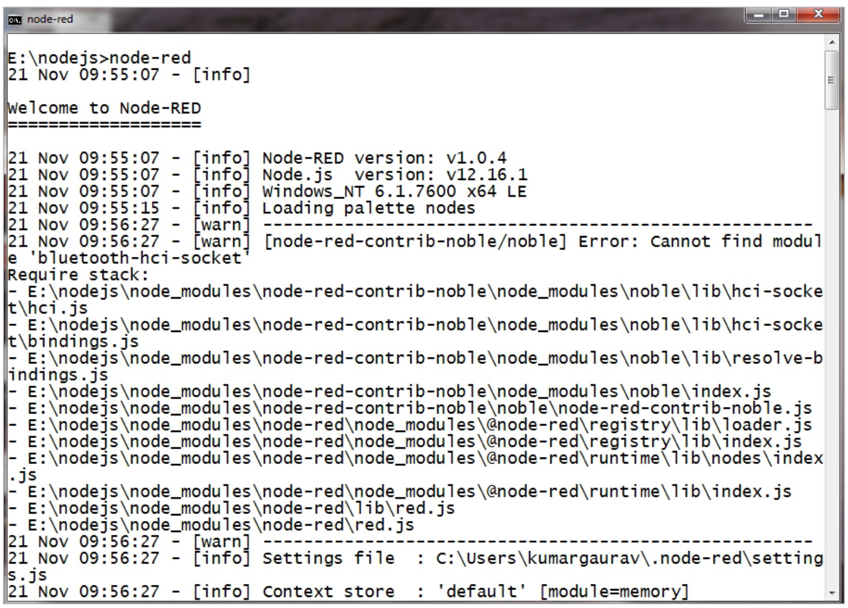


Figure 3: Installation options for Node-RED (Source: *nodered.org*)



```
node-red
E:\nodejs>node-red
21 Nov 09:55:07 - [info]

Welcome to Node-RED
===================

21 Nov 09:55:07 - [info] Node-RED version: v1.0.4
21 Nov 09:55:07 - [info] Node.js  version: v12.16.1
21 Nov 09:55:07 - [info] Windows_NT 6.1.7600 x64 LE
21 Nov 09:55:15 - [info] Loading palette nodes
21 Nov 09:56:27 - [warn] ------------------------------------------------------
21 Nov 09:56:27 - [warn] [node-red-contrib-noble/noble] Error: Cannot find modul
e 'bluetooth-hci-socket'
Require stack:
- E:\nodejs\node_modules\node-red-contrib-noble\node_modules\noble\lib\hci-socke
t\hci.js
- E:\nodejs\node_modules\node-red-contrib-noble\node_modules\noble\lib\hci-socke
t\bindings.js
- E:\nodejs\node_modules\node-red-contrib-noble\node_modules\noble\lib\resolve-b
indings.js
- E:\nodejs\node_modules\node-red-contrib-noble\node_modules\noble\index.js
- E:\nodejs\node_modules\node-red-contrib-noble\noble\node-red-contrib-noble.js
- E:\nodejs\node_modules\node-red\node_modules\@node-red\registry\lib\loader.js
- E:\nodejs\node_modules\node-red\node_modules\@node-red\registry\lib\index.js
- E:\nodejs\node_modules\node-red\node_modules\@node-red\runtime\lib\nodes\index
.js
- E:\nodejs\node_modules\node-red\node_modules\@node-red\runtime\lib\index.js
- E:\nodejs\node_modules\node-red\lib\red.js
- E:\nodejs\node_modules\node-red\red.js
21 Nov 09:56:27 - [warn] ------------------------------------------------------
21 Nov 09:56:27 - [info] Settings file  : C:\Users\kumargaurav\.node-red\setting
s.js
21 Nov 09:56:27 - [info] Context store  : 'default' [module=memory]
```

Figure 4: Starting the Node-RED server